

Homework 2

Instructions:

- Take the acceleration due to gravity to be $g = 9.80 \text{ m/s}^2$
- Express all numerical results to two decimal places

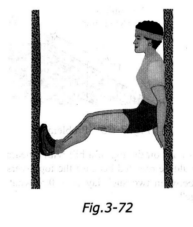
Please submit your answer to the Web Learning by next Thursday, March 19th.

1. Rock Climber in a Chimney

In Fig.3-72, a 55 kg rock climber is climbing in a narrow vertical gap (“chimney”) between two rock walls. The coefficient of static friction between her shoes and the rock is 1.2, and the coefficient of static friction between her back and the rock is 0.80.

She presses against the walls just hard enough so that both her shoes and her back are on the verge of slipping.

- Draw a free-body diagram of the climber.
- What is the magnitude of the horizontal force she exerts on each wall?
- What fraction of her weight is supported by the friction force on her shoes?



2. Three-Mass Pulley System

Three masses are connected by light strings and frictionless pulleys as shown in the Fig.2.35. The upper pulley is fixed, and the lower pulley is movable.

The masses are $m_1 = 200 \text{ g}$, $m_2 = 100 \text{ g}$, $m_3 = 50 \text{ g}$.

- Find the acceleration of each mass. (b) Determine the tensions T_1 and T_2 in the two strings. Assume the strings and pulleys are massless and frictionless.

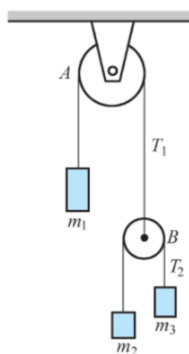
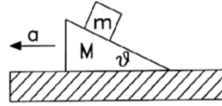


图 2.35

3. Inclined plane on frictionless surface

A small block of mass m is placed on a frictionless inclined plane of mass M which in turn is resting on a frictionless horizontal surface. The inclination angle is θ . Find the horizontal acceleration of the inclined plane due to gravitational force acting on the system. Assume that no outside forces act on the system.



4. Blocks and Pulley on an Accelerating Cart I

What horizontal force must be applied to the cart shown in Figure P5.69 so that the blocks remain stationary relative to the cart? Assume all surfaces, wheels, and pulley are frictionless. (Hint: Note that the force exerted by the string accelerates m_1 .)

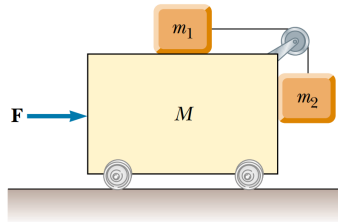


Figure P5.69 Problems 69 and 70.

5. Blocks and Pulley on an Accelerating Cart II

Initially the system of masses shown in Figure P5.69 is held motionless. All surfaces, pulley, and wheels are frictionless. Let the force $\mathbf{F}$ be zero and assume that m_2 can move only vertically. At the instant after the system of masses is released, find

- (a) the tension T in the string,
- (b) the acceleration of m_2 ,
- (c) the acceleration of M , and
- (d) the acceleration of m_1 .

(Note: The pulley accelerates along with the cart.)

6. Bead on a Rotating Circular Hoop

A small bead can slide without friction on a wire bent into a circle of radius 15.0 cm, as in Figure P6.67. The circular wire lies in a vertical plane and rotates steadily about its vertical diameter. The period of rotation is 0.450 s. The position of the bead is described by the angle θ that the radial line from the center of the circle to the bead makes with the vertical.

- (a) At what angle above the lowest point can the bead remain motionless relative to the rotating hoop?
- (b) Repeat the problem if the period of rotation is 0.850 s.

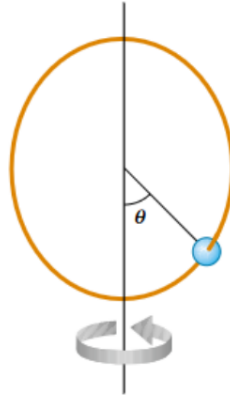


Figure P6.67

7. Pressure Inside a Self-Gravitating Sphere

Consider a spherical body of radius R and uniform density ρ held together only by its own gravity.

- (a) Show that the pressure at the center of the sphere is

$$p = \frac{2}{3}\pi G\rho^2 R^2.$$

- (b) The Sun is mainly composed of hydrogen and has an average density

$$\rho \approx 1.3 \times 10^3 \text{ kg/m}^3$$

and radius

$$R \approx 7.0 \times 10^8 \text{ m}.$$

Estimate the pressure at the center of the Sun and express your answer in atmospheres ($1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$).

8. Mass and string

A mass m whirls around on a string which passes through a ring as shown. Neglect gravity. Initially the mass is distance r_0 from the center and is revolving at angular velocity ω_0 . The string is pulled with constant velocity V starting at $t = 0$ so that the radial distance to the mass decreases. Draw a force diagram and obtain a differential equation for ω . This equation is quite simple and can be solved either by inspection or by formal integration. Find.

- (a) $\omega(t)$
 (b) The force needed to pull the string.

